

S3.1 Unit description

Combinations of random variables; sampling; estimation, confidence intervals and tests; goodness of fit and contingency tables; regression and correlation

S3.2 Assessment information

Prerequisites

A knowledge of the specifications for S1 and S2 and their prerequisites and associated formulae is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. The paper will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: +, −, ×, ÷, π, x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , $x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

$aX \pm bY \sim N(a\mu_x \pm b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2)$ where X and Y are independent and $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$.

1 Combinations of random variables

What students need to learn:

Distribution of linear combinations of independent Normal random variables.

If $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$

independently, then

$$aX \pm bY \sim N(a\mu_x \pm b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2).$$

No proofs required.

2 Sampling

What students need to learn:

Methods for collecting data. Simple random sampling. Use of random numbers for sampling.

Other methods of sampling: stratified, systematic, quota.

The circumstances in which they might be used. Their advantages and disadvantages.

3 Estimation, confidence intervals and tests

What students need to learn:

Concepts of standard error, estimator, bias.

The sample mean, $\bar{x}$, and the sample variance,

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \text{ as}$$

unbiased estimates of the corresponding population parameters.

The distribution of the sample mean $\bar{X}$.

$\bar{X}$ has mean μ and variance $\frac{\sigma^2}{n}$.

If $X \sim N(\mu, \sigma^2)$ then $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

Concept of a confidence interval and its interpretation.

Confidence limits for a Normal mean, with variance known.

Hypothesis tests for the mean of a Normal distribution with variance known.

Use of Central Limit theorem to extend hypothesis tests and confidence intervals to samples from non-Normal distributions. Use of large sample results to extend to the case in which the variance is unknown.

Hypothesis test for the difference between the means of two Normal distributions with variances known.

Use of large sample results to extend to the case in which the population variances are unknown.

Link with hypothesis tests.

Students will be expected to know how to apply the Normal distribution and use the standard error and obtain confidence intervals for the mean, rather than be concerned with any theoretical derivations.

Use of $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim N(0, 1)$.

$\frac{\bar{X} - \mu}{S/\sqrt{n}}$ can be treated as $N(0, 1)$ when n is large. A knowledge of the t -distribution is not required.

Use of
$$\frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}} \sim N(0, 1).$$

Use of
$$\frac{(\bar{X} - \bar{Y}) - (\mu_x - \mu_y)}{\sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}}} \sim N(0, 1).$$

A knowledge of the t -distribution is not required.

4 Goodness of fit and contingency tables

What students need to learn:

The null and alternative hypotheses. The use of $\sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$ as an approximate χ^2 statistic.

Degrees of freedom.

Applications to include the discrete uniform, binomial, Normal, Poisson and continuous uniform (rectangular) distributions. Lengthy calculations will not be required.

Students will be expected to determine the degrees of freedom when one or more parameters are estimated from the data. Cells should be combined when $E_i < 5$. Yates' correction is not required.

5 Regression and correlation

What students need to learn:

Spearman's rank correlation coefficient, its use, interpretation and limitations.

Testing the hypothesis that a correlation is zero.

Numerical questions involving ties will not be set. Some understanding of how to deal with ties will be expected.

Use of tables for Spearman's and product moment correlation coefficients.